Energy-space absorption: the mathematics

Three questions: the change of variables ε → *E*, the momentum transfer *q* and the dirty approximation, and the two gap values Δ and Δ′

Model and notation of HHM, arXiv:1801.08988 ($\Delta_{ks} = \Delta^0 - \Delta^s\varepsilon_{ks}/4t$, Eq. 37; $E_{ks} = \sqrt{(\xi^2 + |\Delta_{ks}|^2)}$, Eq. 26); absorption formalism of arXiv:2607.02812. Equations are numbered (1), (2), …; "state *a*" = (**k**,s), "state *b*" = (**k**′,s′).

> **Answers in one line each.** (1) The sum over states can be written over ε or over E: it is the *same integral* (two branches, Jacobian $E/\kappa\ell$, limits $E_{\min} \le E \le E_{\max}$). What is *not* equivalent is imposing energy conservation on ε instead of E. (2) q enters only through the Kronecker delta $\delta_{\mathbf{k},\mathbf{k}'+\mathbf{q}}$ of the perturbation; the dirty approximation replaces it by its average $1/N_k$, so $\mathbf{k}$ and $\mathbf{k}'$ are summed independently and q disappears (it survives only in the validity condition $q\ell \ll 1$). (3) There is one gap function $\bar{\Delta}(\varepsilon)$; Δ and Δ' are its values at the band energies of the two states. In energy space each state has two branches, $\Delta_\sigma(E) = \Delta^* - \sigma b\ell/\kappa$, and $\xi\xi' + \Delta\Delta' = E_{\min}^2 + \sigma\sigma'\ell\ell'$ for $\eta = -1$.

0. Notation

$\bar{\Delta}(\varepsilon) = \Delta^0 - b\,\varepsilon$ with $b = \Delta^s/4t$ (HHM Eq. 37), $\xi = \varepsilon - \mu$, so that $\bar{\Delta} = \Delta_F - b\xi$ with $\Delta_F = \Delta^0 - b\mu$. Put $\kappa = \sqrt{1+b^2}$, $E_{\min} = |\Delta_F|/\kappa$, $\varepsilon^* = \mu + b\Delta_F/\kappa^2$, $\Delta^* = \Delta_F/\kappa^2$, $\xi^* = \varepsilon^* - \mu = b\Delta^*$. $N(\varepsilon) = N_k^{-1}\sum_{\mathbf{k}s}\delta(\varepsilon - \varepsilon_{\mathbf{k}s})$ is the normal density of states of both helicity bands ($\int N\,d\varepsilon = 2$, zero for $|\varepsilon| > W$), so $\sum_{\mathbf{k}s} F(\varepsilon_{\mathbf{k}s}) = N_k\int d\varepsilon\, N(\varepsilon)F(\varepsilon)$. The quasiparticle energy depends on $(\mathbf{k}, s)$ only through ε (HHM Eq. 26 with $|\Delta_{\mathbf{k}s}| = |\bar{\Delta}_{\mathbf{k}s}|$):

$$E(\varepsilon)^2 = \xi^2 + \bar{\Delta}(\varepsilon)^2 = (1+b^2)\xi^2 - 2b\Delta_F\xi + \Delta_F^2 = E_{\min}^2 + \kappa^2(\varepsilon - \varepsilon^*)^2 \tag{1}$$

1. Integral over ε or over *E*?

Step 1 — the map $\varepsilon \mapsto E$ is two-to-one.

By (1), $E \ge E_{\min}$ with the minimum at ε^*. For every $E > E_{\min}$ there are exactly two band energies, one on each side of ε^*:

$$\varepsilon_\sigma(E) = \varepsilon^* + \sigma\frac{\ell}{\kappa}, \qquad \ell \equiv \sqrt{E^2 - E_{\min}^2}, \qquad \sigma = \pm 1, \qquad \left|\frac{dE}{d\varepsilon}\right|_{\varepsilon_\sigma} = \frac{\kappa^2|\varepsilon - \varepsilon^*|}{E} = \frac{\kappa\,\ell}{E} \tag{2}$$

Step 2 — insert $1 = \int_{E_{\min}}^{\infty} dE\,\delta(E - E(\varepsilon))$.

With $\delta(E - E(\varepsilon)) = \sum_\sigma \delta(\varepsilon - \varepsilon_\sigma(E))/|dE/d\varepsilon|_{\varepsilon_\sigma}$:

$$\sum_{\mathbf{k}s} F(\varepsilon_{\mathbf{k}s}) = N_k\int d\varepsilon\, N(\varepsilon)F(\varepsilon) = N_k\int_{E_{\min}}^{E_{\max}} dE \sum_{\sigma\,=\,\pm} \nu_\sigma(E)\,F\left(\,\varepsilon_\sigma(E)\,\right), \qquad \nu_\sigma(E) = N\left(\,\varepsilon_\sigma(E)\,\right)\frac{E}{\kappa\sqrt{E^2 - E_{\min}^2}} \tag{3}$$

This is an identity (a change of variables), not an approximation. The factor $E/\kappa\ell$ is the Jacobian $|d\varepsilon/dE|$; it diverges like $(E - E_{\min})^{-1/2}$ at the lower limit, which is integrable (it is the BCS density-of-states peak) and absent in the ε-form, where $dE = (\kappa\ell/E)\,d\varepsilon \to 0$ cancels it. **Limits:** the lower limit $E_{\min}$ is the minimum of (1). The upper limit is ∞ only for an unbounded band; the lattice band has $N(\varepsilon) = 0$ for $|\varepsilon| > W$, so $\nu_\sigma(E) = 0$ whenever $|\varepsilon_\sigma(E)| > W$ (the band test), i.e. effectively $E \le E_{\max} = \max\{E(-W), E(W)\}$ ($E_{\min} = 0.578\,t$, $E_{\max} = 7.89\,t$ in the example of the code review).

Step 3 — the absorbed power.

The golden rule with a constant matrix element B (Sec. 2) gives, for quasiparticle scattering (a = upper state, $E_a = E_b + \hbar\omega$) and for pair creation ($E_a + E_b = \hbar\omega$, b the lower member, $E_b < E_a$),

$$W_{sc} = 2\pi\omega B^2\sum_{a,\,b} C_{ab}\left[\,f(E_b) - f(E_a)\,\right]\delta(E_a - E_b - \hbar\omega), \qquad W_{pair} = 2\pi\omega B^2\sum_{a,\,b}^{E_b < E_a} C^p_{ab}\left[\,1 - f(E_a) - f(E_b)\,\right]\delta(E_a + E_b - \hbar\omega) \tag{4}$$

with $f(E) = 1/(e^{E/T} + 1)$ (E measured from the condensate: no μ). Applying (3) to both sums and using the delta function to remove E_a:

$$W_{sc} = 2\pi\omega B^2 N_k^2\int_{E_{\min}}^{E_{\max} - \hbar\omega} dE\sum_{\sigma\sigma'}\nu_{\sigma'}(E + \hbar\omega)\,\nu_\sigma(E)\;C_{\sigma'\sigma}(E + \hbar\omega, E)\left[\,f(E) - f(E + \hbar\omega)\,\right] \tag{5}$$

$$W_{pair} = 2\pi\omega B^2 N_k^2\int_{E_{\min}}^{\hbar\omega/2} dE\sum_{\sigma\sigma'}\nu_\sigma(E)\,\nu_{\sigma'}(\hbar\omega - E)\;C^p_{\sigma\sigma'}(E, \hbar\omega - E)\left[\,1 - f(E) - f(\hbar\omega - E)\,\right] \tag{6}$$

In (5) the energy conservation $\delta(E_a - E_b - \hbar\omega)$ and the occupations act on the quasiparticle energies; the sums over σ, σ' run over the admissible branches. **The ε-form used in the code is (5)–(6) written with $\int d\varepsilon_b N(\varepsilon_b)$ instead of $\int dE\sum_\sigma \nu_\sigma(E)$: by (3) the two are**

equal. Numerically: (3) holds to 7×10^{-5} – 1×10^{-3} for three test functions, and the literal E-integrals (5)–(6) agree with the ε-code to 1×10^{-4} (scattering) and 5×10^{-4} (pair creation).

> **What is not equivalent.** The formula of the old code is $\int d\varepsilon\, N(\varepsilon)N(\varepsilon+\hbar\omega)\,\tilde{C}\,[f_N(\varepsilon)-f_N(\varepsilon+\hbar\omega)]$: the delta function is $\delta(\varepsilon_a-\varepsilon_b-\hbar\omega)$ and the occupation is the normal-state f_N. Since $E(\varepsilon)$ in (1) is nonlinear and two-to-one, no change of variables turns $\delta(\varepsilon_a-\varepsilon_b-\hbar\omega)$ into $\delta(E_a-E_b-\hbar\omega)$: it is a different integral, not a different parametrisation of (5). **Check:** for $b=0$ and a flat $N(\varepsilon)=N_0$ one has $\nu_\sigma(E)=N_0E/\ell$ and (5) gives the textbook $W_s/W_n\to 2f(\Delta)$ as $\omega\to 0$ (verified numerically to 10^{-4}).

2. Where is q? What is the dirty approximation?

Step 1 — where q comes from.

A sound wave of wavevector $\mathbf{q}$ and frequency ω couples to the electrons through $H'(t)=\widehat{H}'e^{-i\omega t}+\mathrm{h.c.}$ with

$$\widehat{H}'=\sum_{\mathbf{k}\mathbf{k}'}\sum_{ss'}B(\mathbf{k}s|\mathbf{k}'s')\,a^\dagger_{\mathbf{k}s}a_{\mathbf{k}'s'},\qquad B(\mathbf{k}s|\mathbf{k}'s')=B_{\mathbf{q}}\,\langle\mathbf{k}s|\mathbf{k}'s'\rangle\,\delta_{\mathbf{k},\mathbf{k}'+\mathbf{q}} \tag{7}$$

($\langle\mathbf{k}s|\mathbf{k}'s'\rangle$ is the helicity-spinor overlap, $\omega=v_s|\mathbf{q}|$.) The quasiparticle matrix element is $M_{ab}=B(a|b)\,[u_au_b-\eta\,v_av_b]$, so $|M_{ab}|^2=|B(a|b)|^2\,C_{ab}$ with $C_{ab}=\frac{1}{2}[1+\xi\xi'/EE'-\eta\bar{\Delta}\bar{\Delta}'/EE']$. **Clean limit** (momentum conserved): the Kronecker delta removes the sum over $\mathbf{k}'$, and (4) becomes

$$W_{sc}(\mathbf{q})=2\pi\omega|B_{\mathbf{q}}|^2\sum_{\mathbf{k}ss'}\left|\langle\mathbf{k}s|\mathbf{k}-\mathbf{q},s'\rangle\right|^2C(\mathbf{k}s;\mathbf{k}-\mathbf{q},s')\left[f(E_{\mathbf{k}-\mathbf{q},s'})-f(E_{\mathbf{k}s})\right]\delta\left(E_{\mathbf{k}s}-E_{\mathbf{k}-\mathbf{q},s'}-\hbar\omega\right) \tag{8}$$

Here q appears in three places: in the arguments $\mathbf{k}-\mathbf{q}$ of $\varepsilon, E, \bar{\Delta}, u, v$ (hence in C), in the helicity overlap, and in the energy condition; for small q, $E_{\mathbf{k}s}-E_{\mathbf{k}-\mathbf{q},s}\approx\hbar\,\mathbf{v}_g(\mathbf{k})\cdot\mathbf{q}$, and with $\omega=v_sq$ the resonance condition is $\mathbf{v}_g\cdot\hat{\mathbf{q}}=v_s$. This is a $\mathbf{k}$-resolved calculation, not an energy-space one.

Step 2 — the dirty approximation.

Statement. The paper averages B over the angles of $\mathbf{k},\mathbf{k}'$ and the spin indices and treats it as a constant (following its reference to Tinkham, Ch. 3); it does not use the word "dirty", which is my name for the physical situation in which such an average is justified. Heuristically: with elastic impurity scattering of mean free path ℓ_{mfp} the states are no longer plane waves, and for $q\,\ell_{mfp}\ll 1$ (and $\ell_{mfp}\ll\xi_0$, the usual dirty limit) the phonon cannot select $\mathbf{k}'=\mathbf{k}-\mathbf{q}$. Mathematically the approximation replaces the Kronecker delta in (7) by its average over $\mathbf{q}$, $N_k^{-1}\sum_{\mathbf{q}}\delta_{\mathbf{k},\mathbf{k}'+\mathbf{q}}=1/N_k$, i.e. the matrix element becomes independent of $\mathbf{k},\mathbf{k}'$:

$$\sum_{\mathbf{k}\mathbf{k}'}|B_{\mathbf{q}}|^2\delta_{\mathbf{k},\mathbf{k}'+\mathbf{q}}\,F(\mathbf{k},\mathbf{k}')\longrightarrow\frac{|B_{\mathbf{q}}|^2}{N_k}\sum_{\mathbf{k}\mathbf{k}'}F(\mathbf{k},\mathbf{k}')=\frac{|B_{\mathbf{q}}|^2}{N_k}\sum_{\mathbf{q}'}\sum_{\mathbf{k}}F(\mathbf{k},\mathbf{k}-\mathbf{q}') \tag{9}$$

The right-hand side is the clean result (8) summed over *all* momentum transfers $\mathbf{q}'$ with equal weight. In this model F depends on $(\mathbf{k},s)$ and $(\mathbf{k}',s')$ only through ε,ε' (the coherence factor C and the occupations) and through the helicity overlap. The overlap averages to $\frac{1}{2}$ over each constant-energy contour, because $\hat{\mathbf{g}}_{-\mathbf{k}}=-\hat{\mathbf{g}}_{\mathbf{k}}$ while $\varepsilon_{\mathbf{k}s}$ is even in $\mathbf{k}$:

$$\left\langle\tfrac{1}{2}\left(1+ss'\,\hat{\mathbf{g}}_{\mathbf{k}}\cdot\hat{\mathbf{g}}_{\mathbf{k}'}\right)\right\rangle_{\varepsilon,\varepsilon'}=\tfrac{1}{2}\left(1+ss'\,\langle\hat{\mathbf{g}}_{\mathbf{k}}\rangle_{\varepsilon}\cdot\langle\hat{\mathbf{g}}_{\mathbf{k}'}\rangle_{\varepsilon'}\right)=\tfrac{1}{2} \tag{10}$$

Therefore the sums factorise into energy integrals, with $B^2=|B_{\mathbf{q}}|^2/2N_k$:

$$W=2\pi\omega B^2N_k^2\int d\varepsilon\,d\varepsilon'\,N(\varepsilon)N(\varepsilon')\,C(\varepsilon,\varepsilon')\left[f(E')-f(E)\right]\delta(E-E'-\hbar\omega) \tag{11}$$

which is (5) after Step 2 of Sec. 1, and which has the same B^2 in W_N, so B^2 cancels in W/W_N. **Where is q now?** In no formula: it has been averaged over. The physical value $q=\omega/v_s$ appears only in the validity condition $q\ell_{mfp}\ll 1$; a number such as $qa\approx 43$ for $\hbar\omega=0.7t$ is a statement about that condition (and about the lattice zone), not an input. In the all-pairs sum the absorbed weight is spread over all momentum transfers: 10 %, 50 % and 90 % of it lies at $|q|a<0.67, 1.85, 2.91$ (zone corner 4.44).

> **What this is, and is not.** (9) is an *assumption* of the paper's formalism; I did not derive its range of validity for the Rashba model. If you want a fixed $\mathbf{q}$ (clean limit, $q\ell_{mfp}\gg 1$) you need (8), which is $\mathbf{k}$-resolved. One supporting fact: the pairing kernel of HHM is $V_{ss'}\propto U-2(\Delta t/t)(\varepsilon_{\mathbf{k}s}+\varepsilon_{\mathbf{k}'s'})$ (Eq. 20 with $8\Delta t\,s_{\mathbf{k}}+4(\Delta t/t)V_{SO}sR_{\mathbf{k}}=-2(\Delta t/t)\varepsilon_{\mathbf{k}s}$), so all inputs depend on a state only through its band energy: energy is the natural label once momentum is not conserved.

3. Two gap values, Δ and Δ′

There is **one** gap function, $\bar{\Delta}(\varepsilon) = \Delta_{_F} - b\xi$ (HHM Eq. 37). In the coherence factor $C = \frac{1}{2}[1 + \xi\xi^{'}/EE^{'} - \eta\Delta\Delta^{'}/EE^{'}]$ the unprimed quantities belong to state a and the primed ones to state b: $\Delta = \bar{\Delta}(\varepsilon_a)$, $\Delta^{'} = \bar{\Delta}(\varepsilon_b)$ — the same function at two arguments. (If you mean the complex $\Delta_{\mathbf{k}s} = s\, e^{-i\theta_{\mathbf{k}}}\bar{\Delta}_{\mathbf{k}s}$: $|\Delta_{\mathbf{k}s}| = |\bar{\Delta}_{\mathbf{k}s}|$, and the phase cancels against the time-reversal phase of the helicity vertex, see the helicity note.)

Step 1 — in energy space each state has two branches.

Evaluate $\xi = \varepsilon - \mu$ and $\bar{\Delta} = \Delta_{_F} - b\xi$ at the roots (2): with $\xi_{_\sigma} = \xi^{^*} + \sigma\ell/\kappa$ and $\xi^{^*} = b\Delta^{^*}$,

$$\xi_{_\sigma}(E) = \xi^{^*} + \sigma\frac{\ell}{\kappa}, \qquad \Delta_{_\sigma}(E) = \bar{\Delta}\,(\,\varepsilon_{_\sigma}(E)\,) \;= \Delta_{_F} - b\xi_{_\sigma} = \Delta^{^*} - \sigma\frac{b\,\ell}{\kappa}, \qquad \xi_{_\sigma}^2 + \Delta_{_\sigma}^2 = E_{\min}^2 + \ell^2 = E^2 \tag{12}$$

(The cross term $2\sigma(\ell/\kappa)(\xi^{^*} - b\Delta^{^*})$ in $\xi_{_\sigma}^2 + \Delta_{_\sigma}^2$ vanishes, and $\xi^{^{*2}} + \Delta^{^{*2}} = \kappa^2\Delta^{^{*2}} = E_{\min}^2$.) For energies $E_a = E_b + \hbar\omega$ there are four combinations (σ_a, σ_b), each with its own pair (ξ, Δ), $(\xi^{'}, \Delta^{'})$ and weight $\nu_{_{\sigma_a}}(E_a)\nu_{_{\sigma_b}}(E_b)$; a branch with $|\varepsilon_{_\sigma}| > W$ is dropped.

Step 2 — the product of the two gaps in closed form.

Multiplying the branch values of (12) for the two states (ℓ_a, ℓ_b their ℓ's):

$$\xi_{_a}\xi_{_b} + \Delta_{_a}\Delta_{_b} = E_{\min}^2 + \sigma_{_a}\sigma_{_b}\,\ell_{_a}\ell_{_b} \tag{13}$$

$$\xi_{_a}\xi_{_b} - \Delta_{_a}\Delta_{_b} = -(1-b^2)\Delta^{^{*2}} + \frac{2b\Delta^{^*}}{\kappa}\,(\,\sigma_{_a}\ell_{_a} + \sigma_{_b}\ell_{_b}\,) \;+ \sigma_{_a}\sigma_{_b}\,\frac{(1-b^2)}{\kappa^2}\,\ell_{_a}\ell_{_b} \tag{14}$$

Hence the coherence factors of (5) and (6) are explicit functions of $E_a, E_b, \sigma_a, \sigma_b$ only:

$$C_{\sigma_{_a}\sigma_{_b}}^{(\eta\,=\,-1)} = \frac{1}{2}\left[1 + \frac{E_{\min}^2 + \sigma_{_a}\sigma_{_b}\ell_{_a}\ell_{_b}}{E_{_a}E_{_b}}\right], \qquad C_{\sigma_{_a}\sigma_{_b}}^{(\eta\,=\,+1)} = \frac{1}{2}\left[1 + \frac{\xi_{_a}\xi_{_b} - \Delta_{_a}\Delta_{_b}}{E_{_a}E_{_b}}\right] \text{ with (14)}, \qquad C^{p} = \frac{1}{2}\left[1 - \frac{\xi_{_a}\xi_{_b} - \eta\Delta_{_a}\Delta_{_b}}{E_{_a}E_{_b}}\right] \tag{15}$$

BCS limit $b \to 0$: $\xi^{^*} = 0$, $\Delta^{^*} = \Delta_{_F} = \Delta$, $\kappa = 1$, $E_{\min} = |\Delta|$, $\xi_{_\sigma} = \sigma\ell$, and (13), (14) reduce to $\Delta^2 + \sigma_{_a}\sigma_{_b}\ell_{_a}\ell_{_b}$ and $-\Delta^2 + \sigma_{_a}\sigma_{_b}\ell_{_a}\ell_{_b}$: the standard case-II and case-I coherence factors. For $b \neq 0$ the extra terms are exactly the particle–hole asymmetry of the gap. **Check:** (12)–(14) hold to 1×10^{-14} over 20000 random $(E_a, E_b, \sigma_a, \sigma_b)$.

4. Result

Collecting (5), (6), (15) and the normal-state absorption with its own chemical potential $\mu_{_N}(T)$ at the same filling:

$$\frac{W}{W_N} = \frac{W_{sc} + W_{pair}}{W_N}, \qquad W_N = 2\pi\omega B^2 N_k^2 \int d\varepsilon\, N(\varepsilon)N(\varepsilon + \hbar\omega)\,[\,f_{_N}(\varepsilon) - f_{_N}(\varepsilon + \hbar\omega)\,] \tag{16}$$

Inputs: $\Delta^{^0}(T), \Delta^{^s}(T), \mu(T)$ from the self-consistent solution of HHM Eqs. 32, 33, 38 (no Δ_{eff}); the lattice enters only through $N(\varepsilon)$, the band half-width W and (via ε) the helicity bands. Assumptions: constant B and unrestricted $\mathbf{k}, \mathbf{k}^{'}$ sums (Sec. 2). Not covered: fixed q (clean limit), a direction-dependent $B_{\mathbf{q}}$, and the $q \to 0$ electromagnetic interband vertex — these need the $\mathbf{k}$-resolved form (8).

5. The sign of ξ in the coherence factor, and what to expect for W/W$_{\mathbf{N}}$

What the paper does. Its expansion of $[uu^{'} - \eta vv^{'}]^2$ (written with positive square roots, $v \geq 0$) gives $\frac{1}{2}\left(\,1 + \xi\xi^{'}/EE^{'} - \Delta\Delta^{'}/EE^{'}\,\right)$ with the *signed* $\xi\xi^{'}$ — the correct case-I scattering factor, and the paper remarks that this term cannot be dropped. Two later steps spoil it: (i) the final W carries $+\eta\Delta\Delta^{'}$, whereas $(uu^{'} - \eta vv^{'})^2$ gives $-\eta\Delta\Delta^{'}$; (ii) "in terms of E" the paper replaces $\varepsilon - \mu$ by $\sqrt{E^2 - \Delta^2}$. Squaring $E^2 = \xi^2 + \Delta^2$ erases the sign of ξ, and taking the positive root puts every state on the electron side: $\xi\xi^{'} \to |\xi||\xi^{'}|$.

Why the sign of ξ cannot be dropped.

Write the Bogoliubov coefficients as $u = \cos\varphi$, $v = \sin\varphi$, with 2φ the polar angle of the vector $(\xi, \bar{\Delta})$ (both components signed):

$$\cos 2\varphi = \frac{\xi}{E}, \qquad \sin 2\varphi = \frac{\bar{\Delta}}{E}, \qquad uu^{'} - \eta vv^{'} = \cos(\varphi + \eta\varphi^{'}), \qquad uv^{'} + \eta u^{'}v = \sin(\varphi^{'} + \eta\varphi) \tag{17}$$

Hence, for scattering and for pair creation,

$$C = \cos^2(\varphi + \eta\varphi^{'}) = \frac{1}{2}\left[1 + \frac{\xi\xi^{'} - \eta\bar{\Delta}\bar{\Delta}^{'}}{EE^{'}}\right], \qquad C^{p} = \sin^2(\varphi^{'} + \eta\varphi) = \frac{1}{2}\left[1 - \frac{\xi\xi^{'} - \eta\bar{\Delta}\bar{\Delta}^{'}}{EE^{'}}\right], \qquad C + C^{p} = 1 \tag{18}$$

For $\eta = +1$ (ultrasound) the combination is $\xi\xi^{'} - \bar{\Delta}\bar{\Delta}^{'}$ with $\xi, \bar{\Delta}$ signed. The sign of ξ is the electron/hole label: for $\bar{\Delta} \to 0$, $\varphi = 0$ ($\xi > 0$) or $\pi/2$ ($\xi < 0$), so (18) gives $C = 1$ for two states of the same type and $C = 0$ for opposite types (a scalar vertex cannot turn an electron into a hole), and C^{p} is the complement — opposite-type pairs are created, not scattered. With $|\xi||\xi^{'}|$ one gets $C = 1$ for *every* pair at $\bar{\Delta} = 0$ and no pair creation. **Sum rule at** $\bar{\Delta} \to 0$ (scattering + pair creation must reproduce the normal-state absorption, $T/T_c = 0.7$): signed $\xi\xi^{'}$: 0.3507 + 0.6502 = 1.0008; with $|\xi||\xi^{'}|$: 0.6436 + 0.0000 = 0.6436.

Does the choice change the behaviour of W/W_N?

In the paper's (and the repo's) formula the bracket is $1 + n \cdot n'$ with unit vectors $n = (|\xi|, \Delta)/E$, $n' = (|\xi'|, \eta\Delta')/E'$, hence lies in $[0, 2]$; W_N has bracket 2. Therefore $W/W_N \leq 1$ *identically*, whatever the data: for the repo (bracket in [1.4966, 2.0000]) the ratio is the average of $C = \text{bracket}/2 \in [0.75, 1]$ over the thermal window, 0.93–0.96. This bound is a property of the formula, not evidence for it. It disappears as soon as the sign of ξ is kept in the old integrand (band-energy conservation, normal-state f): normalised by its own $\Delta \to 0$ limit, which vanishes for pairs straddling μ, the ratio becomes (T/T$_c$ = 0.1, 0.3, 0.5, 0.7, 0.8): 16.01, 4.96, 2.80, 2.05, 1.74 with $+\eta\Delta^2$, and 1.04, 0.54, 0.53, 0.64, 0.70 with $-\eta\Delta^2$ (repo inputs, $\hbar\omega = 0.7$). That framework cannot be repaired by choosing signs: the pairs straddling μ need quasiparticle energy conservation and the pair-creation factor $C^{\wp}$.

In the quasiparticle framework (energy conservation on E, $f(E)$, scattering + pair creation, physical W_N; $V_{SO} = 0.4082$, $\hbar\omega = 0.7$, $\eta = +1$; (1)–(3) include $C^{\wp}$ with the same sign choice, (4) has no pair term as in the paper):

T/T$_c$	2E$_{min}$	(1) signed ξξ′, −ηΔΔ′ correct	(2) |ξ||ξ′|, −ηΔΔ′	(3) signed ξξ′, +ηΔΔ′	(4) |ξ||ξ′|, +ηΔΔ′ paper's final, no pair term
0.3	1.372	0.0017	0.0019	0.0024	0.0026
0.5	1.322	0.0312	0.0358	0.0421	0.0467
0.7	1.156	0.1289	0.1540	0.1685	0.1936
0.9	0.746	0.3221	0.4194	0.3922	0.4895
0.96	0.487	1.1926	1.1771	0.7264	0.6292

Below the pair-breaking onset ($2E_{\min} > \hbar\omega$, $T/T_c < 0.91$) all four stay below 1 but differ by tens of per cent ((4) is 50 % above (1) at $T/T_c = 0.7$ and 0.9); above the onset the sign of the η term and the missing pair channel change the result by a factor 2.

Should $W/W_N < 1$ be expected?

In the correct formalism there is no bound like the one above. Textbook BCS limit (flat N, constant gap, $\mu = 0$, $T \to 0$; bare golden rule, no vertex corrections):

ħω/2Δ	0.9	1.005	1.1	1.3	2.0	3.0	6.0
W$_s$/W$_n$, η = +1 (ultrasound)	0.000	1.556	1.501	1.395	1.211	1.114	1.037
W$_s$/W$_n$, η = −1 (EM)	0.000	0.008	0.140	0.340	0.672	0.833	0.949

For $\hbar\omega < 2\Delta$ only scattering contributes and $W_s/W_n < 1$ ($\to 2f(\Delta)$ as $\omega \to 0$, verified to 10^{-4}). Above 2Δ the pair-creation factor $C^{\wp} = \sin^2(\varphi + \varphi')$ of case I jumps to $\approx \pi/2$ at the threshold (the two BCS edges, $C^{\wp} \to 1$) and decays to 1 from above; for $\eta = -1$ it rises continuously from 0. **In our model** ($V_{SO} = 0.4082$, correct formula, $\eta = +1$): $\hbar\omega = 0.7$: maximum 1.402 at $T/T_c = 0.9828$, below the onset (T/T$_c$ < 0.9137) maximum 0.339; $\hbar\omega = 0.2$: maximum 1.007 at $T/T_c = 0.9947$, below the onset (T/T$_c$ < 0.9939) maximum 0.767; $\hbar\omega = 0.05$: maximum 0.941 at $T/T_c = 0.9995$, below the onset (none reached) maximum 0.941. So $W/W_N < 1$ is expected for $\hbar\omega < 2E_{\min}(T)$, in particular for ultrasound with $\hbar\omega \ll \Delta$ (apart from a window of relative width ~$(\hbar\omega/2\Delta_0)^2$ below T_c); with $\hbar\omega = 0.7t$, comparable to the gap, the bare formalism gives a bump above 1 for $\eta = +1$ near T_c. At $\hbar\omega \gtrsim 2\Delta$ the density-type vertex is also where vertex corrections (gauge invariance) are expected to matter, which are not included.